



CASE STUDY REPORT

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VIRGIN MOBILE KSA BREACH (2020)



Deliverable 1 — Description of the Selected Real-World Incident

1. Introduction

This case study analyzes a real cyber incident targeting **Virgin Mobile Saudi Arabia** in early **2020**. The attack was confirmed by multiple cybersecurity intelligence platforms, including BankInfoSecurity and Hackmageddon. The breached data was later posted for sale on dark-web marketplaces, indicating a successful compromise of internal systems.

2. Overview of the Incident

In May–June 2020, hackers infiltrated Virgin Mobile KSA's internal network, gaining unauthorized access to:

- **Corporate email systems**, and
- **Active Directory (AD)** domain infrastructure.

The attackers exfiltrated sensitive data, including internal documents, employee information, and configuration details. Later, this data appeared on cyber-criminal forums.

3. Attack Classification

The attack falls under:

- **Network Intrusion**
- **Unauthorized Access**
- **Data Exfiltration / Breach**

Deliverable 2 — How the Attack Worked

Technique Used

The primary technique used in this attack was ARP Spoofing combined with Packet Sniffing, following the compromise of internal credentials. After gaining initial access to the internal network, the attackers exploited weaknesses in internal network monitoring and segmentation to intercept traffic and access critical systems.

Steps Taken by the Attackers

1. Initial Network Access

The attackers gained access to Virgin Mobile Saudi Arabia's internal corporate network using compromised user credentials. This access allowed them to connect to internal systems without triggering perimeter security controls.

2. Internal Positioning and Traffic Interception

Once inside the internal LAN, the attackers performed ARP Spoofing by sending forged ARP reply messages to associate their MAC address with the IP address of legitimate internal servers, such as email servers and Active Directory-related systems. This enabled the attackers to place themselves as a man-in-the-middle (MITM) between employees and internal services.

3. Packet Sniffing and Credential Harvesting

After successfully poisoning ARP tables, internal network traffic was redirected through the attackers' systems. Using packet sniffing techniques, the attackers captured unencrypted data, authentication requests, and session information. This allowed them to collect additional credentials and gain deeper visibility into internal communications.

4. Privilege Escalation and Lateral Movement

With harvested credentials and intercepted authentication data, the attackers escalated privileges and moved laterally within the network. They accessed Active Directory Domain Controllers, user accounts, group

policies, and configuration files, significantly expanding their control over the environment.

5. Data Collection and Exfiltration

Sensitive information—including employee data, internal documents, and email archives—was aggregated, compressed, and encrypted. The data was then exfiltrated to external attacker-controlled servers and later offered for sale on dark-web marketplaces.

Why This Attack Was Effective

This attack succeeded due to:

- Lack of internal network segmentation
- Absence of ARP spoofing detection mechanisms
- Limited encryption of internal traffic
- Over-privileged internal accounts

These weaknesses allowed the attackers to intercept traffic, escalate privileges, and extract sensitive data without immediate detection.

Deliverable 3 — Impact Analysis

1. Operational Impact

- Disruption to email communication channels
- Mandatory resets of Active Directory and email credentials
- Increased load on IT and SOC teams

2. Business Impact

- Reputational damage
- Potential financial loss due to remediation
- Increased scrutiny from telecom regulatory entities

3. Security Impact

- Exposure of internal system configurations
- Potential for future attacks using leaked information
- Loss of confidentiality and integrity of stored user data

4. Long-Term Risks

- Social engineering using leaked data
- Credential stuffing against other systems
- Attackers selling access to the network

Deliverable 4 — Security Controls & Recommendations

A. Technical Controls

1. Identity & Access Management

- Mandatory **Multi-Factor Authentication (MFA)**
- Enforced **password rotation every 90 days**
- Strict policies against password reuse

2. Network Security

- Network segmentation (separating AD, email servers, and user networks)
- Deployment of IDS/IPS to detect:
 - ARP spoofing
 - Anomalous internal scanning
 - Rogue devices

3. Email Security

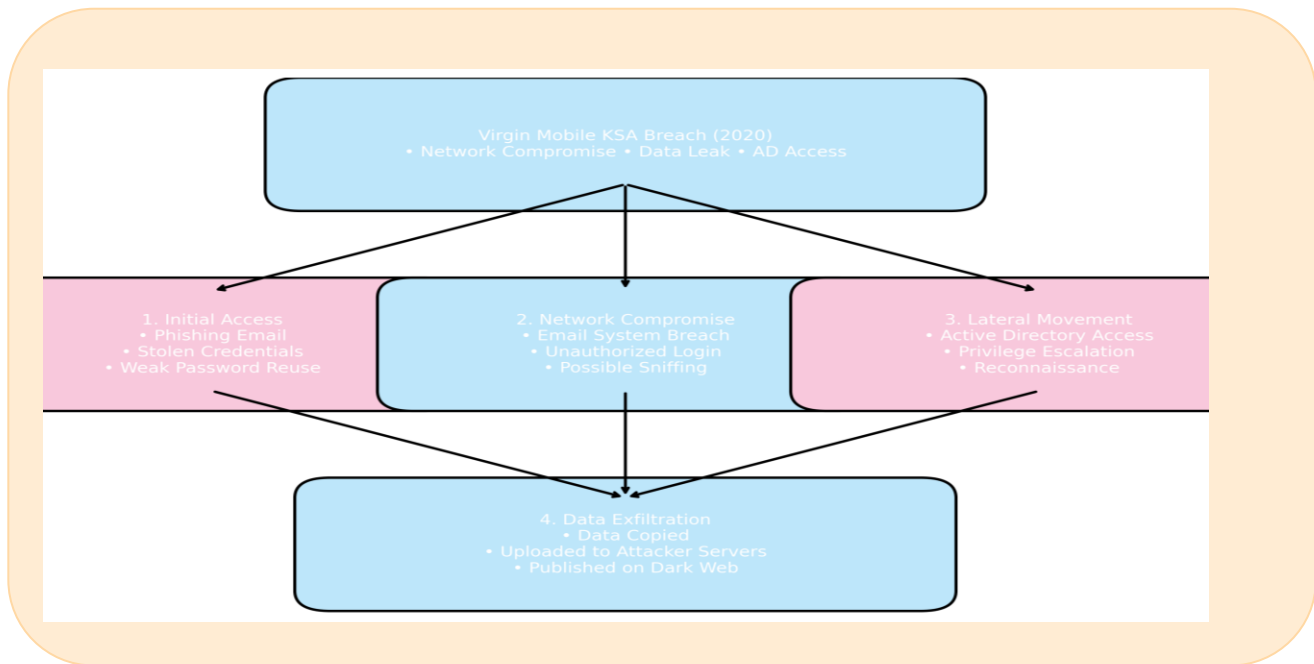
- Advanced phishing protection
- Sandboxing suspicious attachments
- DMARC, DKIM, SPF enforcement

4. Endpoint Protection

- EDR/XDR on servers and workstations
- Privileged Access Workstations (PAWs) for AD admins

5. Data Loss Prevention (DLP)

- Prevents unauthorized data transfer
- Blocks large data exfiltration attempts



Deliverable 5 — Summary Infographic

A clean conceptual map summarizing the incident phases is included:



virgin Mobile Saudi Arabia Timeline :

Conclusion

The Virgin Mobile KSA breach demonstrates how a single point of compromise — such as credential theft — can cascade into a full-scale internal intrusion involving email systems, Active Directory, and eventual data exfiltration.

Even without full public forensic disclosure, the incident aligns with typical multi-stage attack models. This includes potential internal network threats relevant to your research requirements such as ARP spoofing, packet sniffing, and rogue devices.

By understanding each stage of the breach and enforcing preventive controls, organizations can significantly reduce the risk of similar attacks in the future.

References :

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